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Building AI Projects

Starting an AI project

Starting an AI project

- Workflow of projects
- Selecting AI projects
- Organizing data and team for the projects



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Building AI Projects

Workflow of a
machine learning project

Example: Speech recognition



Amazon
Echo / Alexa



Google
Home



Apple
Siri



Baidu
DuerOS

Key steps of a machine learning project

Echo / Alexa

1. Collect data

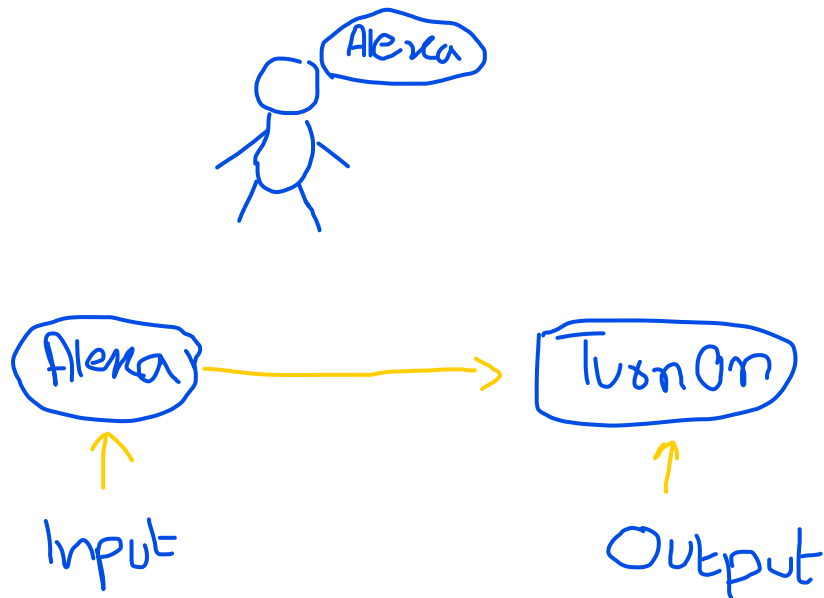
2. Train model

Iterate many times until
good enough

3. Deploy model

Get data back

Maintain / update model





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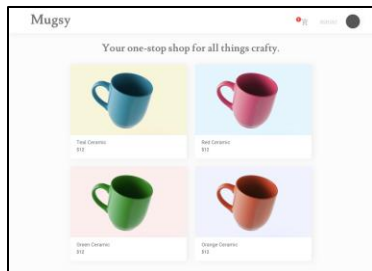
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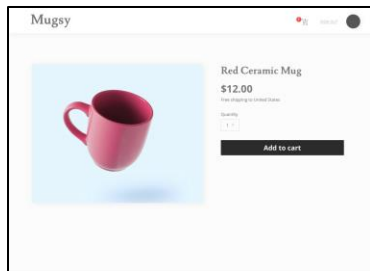
Workflow of a
data science project

Example: Optimizing a sales funnel

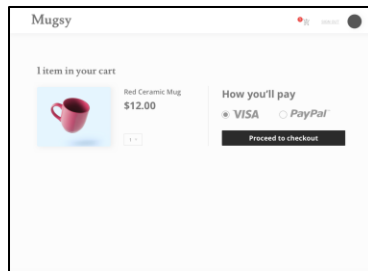
Visit website



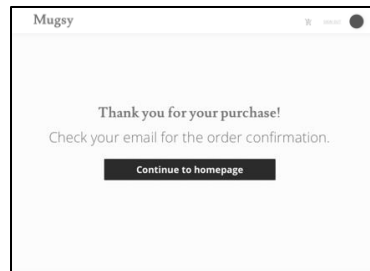
Product page



Shopping cart



Checkout



Key steps of a data science project

Optimizing a sales funnel

1. Collect data

User ID	Country	Time	Webpage
2009	Spain	08:34:30 Jan 5	home.html
2897	USA	13:20:22 May 18	redmug.html
4893	Philippines	22:45:16 Jun 11	mug.html

2. Analyze data

Iterate many times to get good insights

3. Suggest hypotheses/actions

Deploy changes

Re-analyze new data periodically

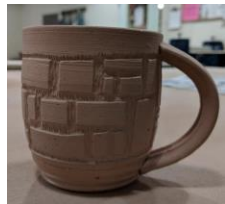
Key steps of a data science project

Manufacturing line

Mix clay



Shape mug



Add glaze



Fire kiln



Final
inspection



Clay Batch #	Supplier	Mixing time (minutes)
001	ClayCo	35
034	GooClay	22
109	BrownStuff	28

1. Collect data

2. Analyze data

Iterate many times to get good insight

3. Suggest hypotheses/actions

Deploy changes

Re-analyze new data periodically

Mug Batch #	Country	Humidity	Temperature in kiln (F)	Duration in kiln (hours)
301	Spain	0.002%	1410°	22
302	USA	0.003%	1520°	24
303	Malaysia	0.002%	1420°	22



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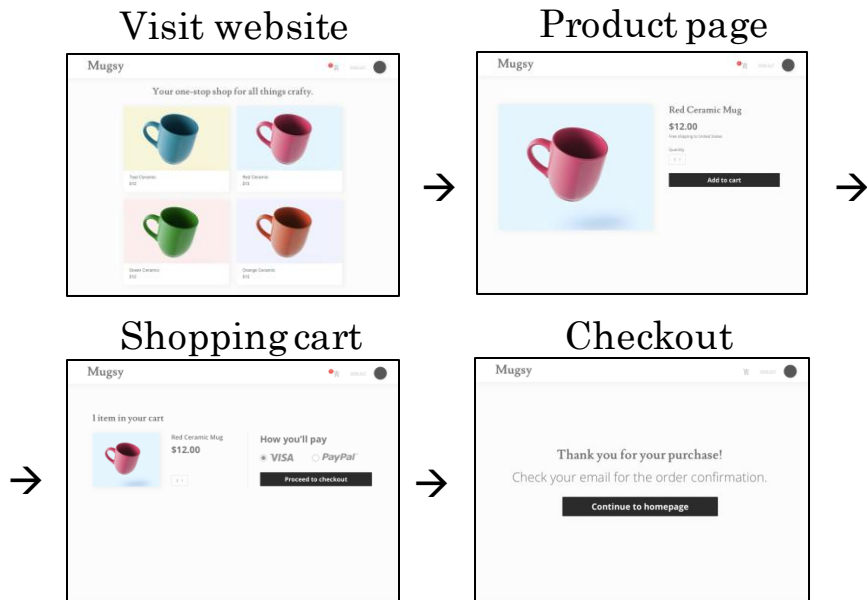
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Building AI Projects

Every job function
needs to learn how to use data

Sales

Data science



Machine learning

Name	Title	Company size	Email	Priority
Tayler	CEO	3050	tay@a..	high
Janet	Manager	230	jan@b..	medium
David	Intern	30	dave@c..	low

Automated lead sorting

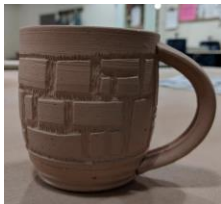
Manufacturing line manager

Data science

Mix clay



Shape mug



Add glaze



Fire kiln



Final inspection



Optimize sales funnel

Machine learning



ok



ok

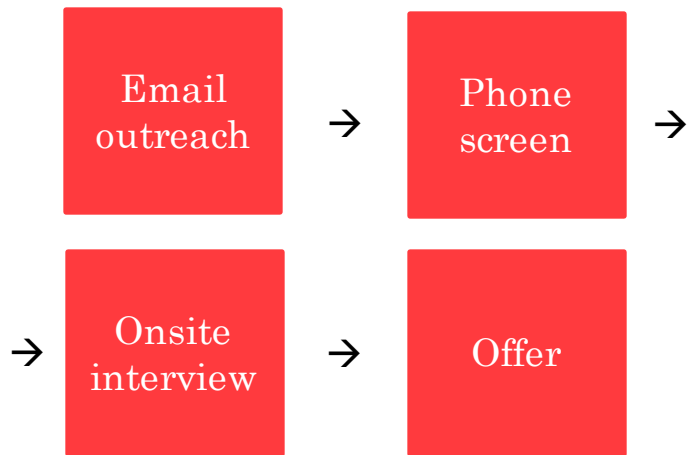


defect

Automated visual inspection

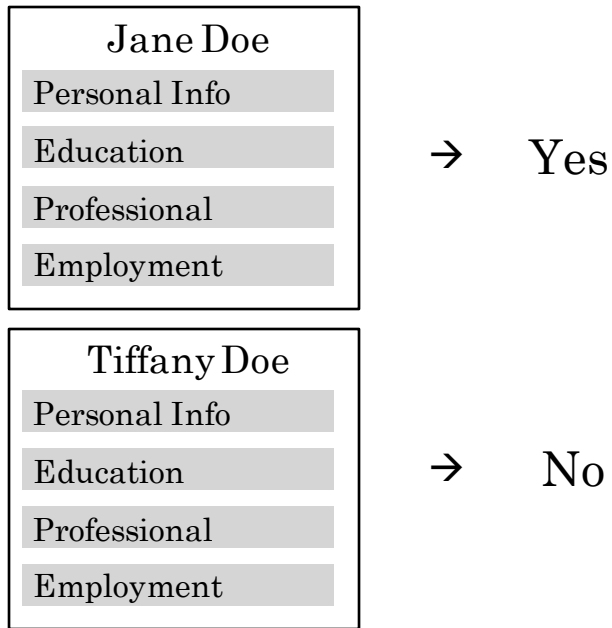
Recruiting

Data science



Optimize recruiting funnel

Machine learning

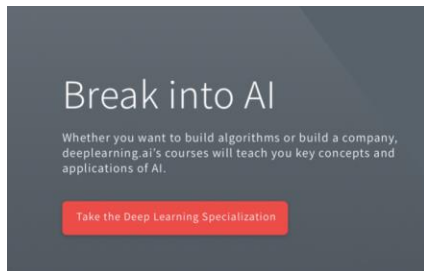


Automated resume screening

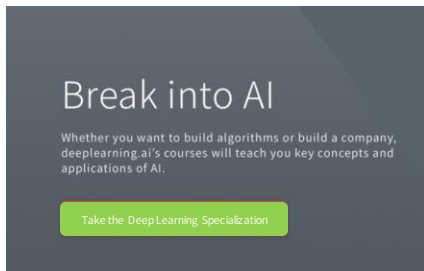


Marketing

Data science



A



B

A/B testing

Machine learning

Recommended for you



Customized product recommendation



Agriculture

Data science



Crop analytics

Machine learning



Precision weed killing



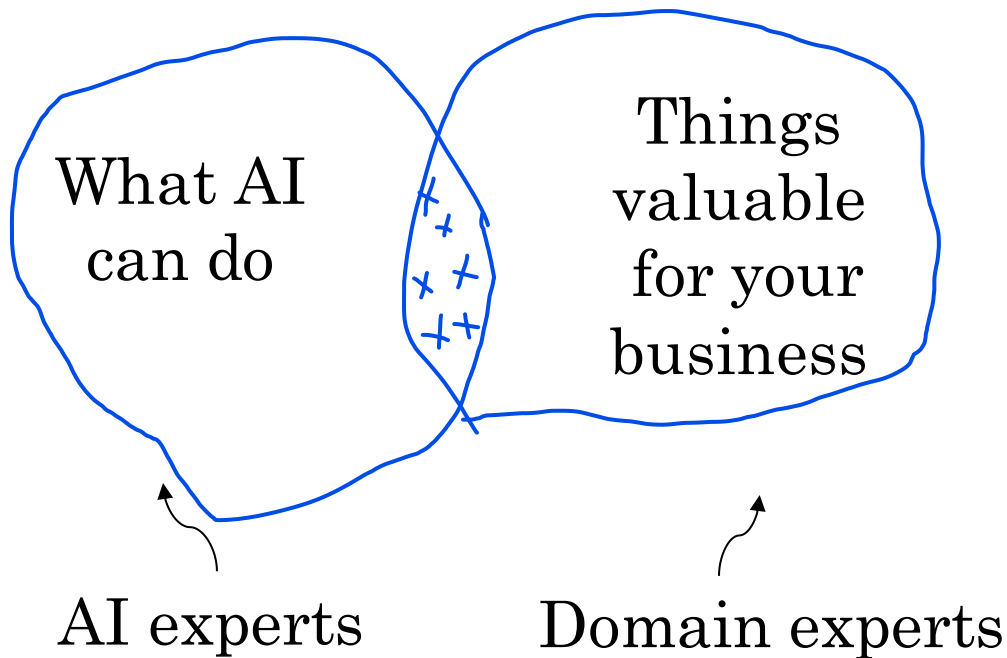
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Building AI Projects

How to choose an AI project I

AI knowledge and domain knowledge



Brainstorming framework

- Think about optimizing tasks rather than automating jobs. E.g., call center routing, radiologists.
- What are the main drivers of business value?
- What are the main points in your business?

You can make progress even without big data

- Having more data almost never hurts.
- Data makes some businesses (like web search) defensible.
- But with small datasets, you might still make progress.





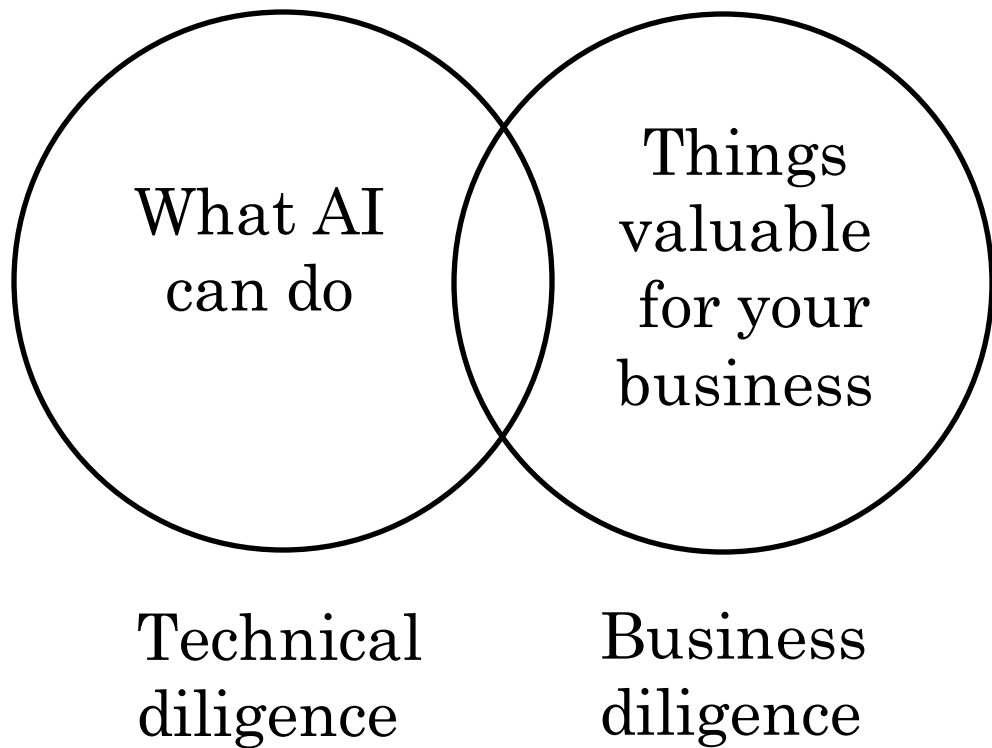
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Building AI Projects

How to choose an AI project II

Due diligence on project



Due diligence on project

Technical diligence

- Can AI system meet desired performance
- How much data is needed
- Engineering timeline

Business diligence

- Lower costs current business
- Increase revenue
- Launch new product or business new business

Build vs. buy

- ML projects can be in-house or outsourced
- DS projects are more commonly in-house
- Some things will be industry standard – avoid building those.



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Building AI Projects

Working with an AI team

Specify your acceptance criteria



ok

Goal: detect defects
with 95% accuracy



ok

Provide AI team a dataset on
which to measure their
performance



defect

How AI teams think about data

Training set



ok



ok



ok

Test set



ok



ok



defect

Pitfall: Expecting 100% accuracy

Test set



ok



ok



ok



defect



ok

- Limitations of ML
- Insufficient data
- Misabeled data
- Ambiguous label



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Building AI Projects

Technical tools for
AI teams (optional)

Open-source frameworks

Machine learning frameworks:

- PyTorch
- TensorFlow
- Hugging Face
- PaddlePaddle
- Scikit-learn
- R

Research publications

- Arxiv

Open source repositories:

- GitHub

CPU vs. GPU

CPU: Computer processor (Central Processing Unit)



GPU: Graphics Processing Unit



Cloud vs. On-premises